Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/27/2007
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Spatial and Temporal Analysis of Gravitropism In Chara Rhizoids		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
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Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Received NCE through 6/30/2007, per A. Chu (9/06)		
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Task Description:	The proposal focuses on the sequence of events of gravisensing in rhizoids of the green alga Chara as a model system for higher plants. The goal is to analyze the sequential and spatial coordination of cellular responses to reorientation. Despite the long-standing correlation between the sedimentation of statoliths and subsequent growth responses, little is known about the transduction of the stimulus, how the positions of the putative sensors are maintained or the steps that lead to differential elongation of either the cell wall of (Chara) rhizoids or an entire cross section of a root or stem. Based on a previously developed microinjection system capable of delivering micron-sized particles (comparable to endogenous statoliths) we propose to use these particles as statoliths and in their oligo(dT)-coated version as tools for extraction of mRNA and subsequent RT-PCR analyses. The magnetic properties of the beads allow their positioning to regions of interest and facilitate the recovery of particles after binding of mRNA. Together with an external magnet the magnetic properties of the beads can also be used to vary the forces acting on artificial statoliths, comparable to gravistimulation with varying forces. Based on this bead-based system and control experiments that will probe various areas of the rhizoid with oligo(dT)-coated glass probes (= solid phase gene extraction, SPGE), we will analyze the distribution and quantity of mRNA's in different zones of the cell during the graviresponse cycle. The proposed work will identify genes and their localization in a single graviresponding cell. The work will result in a novel, stimulus-response analysis that will correlate temporal and spatial information with gene expression patterns during graviresponse. The proposed approach will also discern between primary (= graviresponse) messages and induced secondary (e.g., stress) signals. The experiments will deepen our understanding of gravity effects on a molecular level that is focused on the gravisensing and transduction process. The aim of this project is to enhance our understanding of the nature of perceptions and responses of plants to changes in their environment, especially gravity. The proposed work is aimed at providing fundamental knowledge on plant growth, perception of forces and the sequential and spatial organization of responses to such changes. The project focuses on the effects of gravity, but also includes the perception of external (friction, light, mechanical stress) and internal (changes in viscosity, cellular perturbations) factors which are vital for the development and growth of plants. The project is in line with NASA's quest to improve the quality of life here and extend it to space exploration. It is essential to achieving long-duration human presence in and exploration of space. The proposed work combines novel cellular and molecular approaches to understanding of the perception of and response to the gravity signal and subsequent transduction. The genetic analyses represent not only a new technology but also the first attempt to characterize a valuable and widely studied model system for which almost no genetic data are available. The proposed approach will establish a thorough and, at least for plant systems, novel screening methodology. Specific experiments are designed to follow gene expression over time and (cellular) space. The research will integrate the training of post-docs with molecular, general plant biology and specifically space biology. The project includes many technical aspects (microinjection, genetic analyses, growth studies, hormonal analyses, and microscopy) as well as numerous conceptual facets (spatial and temporal analysis of events and distributions, modeling of gene expression and identification of factors and processes contributing to differential elongation). The potential payoff is high in identifying key regulatory elements for the interaction of physical forces (gravity), auxin and genetic integration and elucidating how these factors are controlled and contribute to the response of plants to gravity. The project will train plant biologists in gravitational biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Aside from the research on gene activation, sequestration and transport, the current work breaks new ground in rapid (< 1 minute) mRNA extraction. Potential future applications may eliminate or reduce the need for destructive tissue biopsies for e.g., cancer or other disease diagnosis.
	The technical aspect of the proposal included the establishment of microprobes that allow rapid, quantitative and sensitive, ideally non-destructive removal of mRNA from living cells. The need for such technology ranges from diagnostic procedures (e.g. cancer) to monitoring purposes (assessment of gene expression as a result of environmental changes). The need to measure mRNA extends from sampling of complete organisms to individual cells at high temporal and spatial resolution. Current small-scale mRNA extractions depend on the removal of the cells/tissue from an organism or preserved specimens. These methods are destructive and do not distinguish between genomic DNA and RNA. Therefore extracted (m)RNA is typically contaminated by extracted cytoplasm, nuclear DNA or other compounds and the required purification leads to loss of especially low-abundant mRNA. The development of Solid Phase Gene Extraction (SPGE) alleviates time consuming and difficult sample purification. SPGE sampling can be achieved using gene specific or generic sequences and is not species specific. After establishing the necessary chemical principles for the preparation of our probes, we demonstrated the versatility and validity of this extraction method by simultaneously profiling nanos and bicoid mRNA in individual Drosophila eggs. SPGE reports previously illustrated distribution profiles of both genes. Its low impact is underscored by the normal development of repeatedly sampled eggs, which eventually developed into adult flies. The technology was also verified in plant systems where the cell wall represents a far greater mechanical impediment than in animal cells. Quantification of different isoforms of actin mRNA in germinating flax seeds linked gene expression to distinct developmental processes. These data demonstrate the universality of SPGE as simple, generic, analytical and diagnostic procedure. Employing considerably less technology than for example laser dissection microscopy, cell-aspiration or microfluidics, SPGE is rapid and can be performed in less than two minutes from extraction to reverse transcription. This technique minimizes contamination, does not require additional purification of the sample or removal of cells, tissue or cytoplasm from the organism. SPGE employs needles that are surface-coated with poly(T) or gene-specific oligonucleotide sequences. mRNA is extracted from an individual cell or tissue by direct hybridization to the probe and results in the removal of a representative mRNA sample. The sampled cells remain alive and may be sampled repeatedly at different times or locations. The hybridized mRNA is reverse transcribed and the amount of generated cDNA is sufficient for multiple real-time PCR analyses using various reporter systems without pre-amplification. To corroborate our method, we studied the mRNA extraction from Drosophila eggs and analyzed gene expression in embryonic plant roots. The distribution of nanos and bicoid mRNA in Drosophila eggs has been investigated extensively over the last decades. During oogenesis nanos and bicoid mRNA's are derived from maternal nurse-cells. Nanos mRNA accumulates in the posterior and bicoid mRNA accumulates in the anterior pole of the egg. This distribution persists until 2 h after fertilization. The translation of nanos and bicoid mRNA establishes protein gradients that determine the body axis and

Task Progress:	the formation of the head and abdomen. We sampled individual eggs (ca. 350 µm long) at four equidistant positions from anterior to posterior and simultaneously and determined the absolute copy number of nanos and bicoid mRNA at the time of extraction using a SYBR-Green based real-time PCR essay. To confirm that SPGE is equally applicable to plant or algal systems, we studied the onset of actin gene expression in germinating flax (Linum usitatissimum) seeds and distribution of actin in Chara contraria internode cells. A patent has been applied for the SPGE technology, which is now in the examination phase. All experiments demonstrate that Solid Phase Gene Extraction is ideally suited to investigate gene expression with high sensitivity. SPGE overcomes current limitations of spatial and temporal gene expression profiling. Initial experiments determined basic physical parameters of our needles including various coating densities and extraction protocols in vitro. The tests were conducted in vitro by incubation of SPGE-probes in 1 µl of flax poly(A) mRNA and after reverse transcription from individual needles was quantified as “copies per sample” (cps) using a dual-labeled fluorescent probe-based real-time PCR assay. Initially glass needles were used and therefore a comparison between the ionic interactions of glass surfaces was compared with specific binding between oligo-dT and poly dA. Unspecific binding of mRNA to the glass surface accounted for less than 0.04 % of the mRNA extracted with oligo dT coated needles under otherwise identical conditions. Maximum mRNA extraction was achieved by coating glass needles with 1 µM oligo-dT10. Hybridization of mRNA to glass probes was half-maximal after 20 s and maximum loading was achieved after 40 s. Although longer incubation times did not increase needle loading, subsequent experiments were performed with a hybridization time of 60 s to optimize reproducibility. The release of hybridized mRNA from the needle was temperature dependent with no mRNA remaining at 70º C or higher, which further corroborates specific mRNA-probe interaction. The ratio between the concentration of mRNA in 1 µl solution and the quantity derived from SPGE probes shows a needle capacity equivalent to mRNA contained in about 1 nL of extracted poly-A-mRNA. Actin profiling in germinating flax roots during the first 48 h of flax seed germination showed markedly different abundance of two actin isoforms between root cap, meristem and elongation zone. These experiments will be continued as soon as funding can be obtained to study the extent of maternal mRNA found in seeds, its longevity and relationship to new transcription events. Related studies on Chara probed the distribution of viscosity in the rhizoids. Aside from essential knowledge of distribution of viscosity we discovered a highly anisotropic distribution of viscosity that differed within narrow regions of the rhizoid. Maximal visoscity was determined to be about 140 cP and the highest difference between lateral and longitudinal viscosity was found in the regions of statoliths. The implications of viscous anisotropy has been discussed in a publication that is in press at this time (9/07).
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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